(10) Patterns (Part I)

- Print STAR Pattern (Here we use Nested loop)

```
*
* *
* * *
* * * *
```

① lines (4)
outer loop → 4 times

② no. of times (i)
inner loop → i times

③ what to print = "*"

code:-

```
import java.util.*;

public class Patterns {
    public static void main (String [] args){

        int s;
        Scanner vj = new Scanner (System.in);
L=1     S=~~1~~2
L=2     S=~~1~~~~2~~3      Sop ("Enter the value");
L=3     S=~~1~~~~2~~34     s = vj.nextInt
L=4     S=~~1~~~~2~~~~3~~~~4~~5
L=5         for(int line=1; line<=s; line++){
                // s line
                for(int star =1; star<= line ; star ++) {
                        Sop ("*");
                }
                Sopln( ); // for changing line
            }
        }
}
```

→ Output:- Enter the value : 6

```
*
* *
* * *
* * * *
* * * * *
* * * * * *
```

□ Print INVERTED - STAR Pattern

```
* * * *
* * *
* *
*
```

① lines (4)
outer loop → 4 times

② inner loop → $n-i+1$

Star $= n-i+1$

Code:-

```
import java.util.*;
public class Pattern{
    public static void main (String [] args){

        int n;
        Scanner vj=new Scanner (System.in);
        S.o.p ("Enter the value:");

        n = vj.nextInt ();
```

```
for(int line=1; line<=n; line++){
        for(int star=1; star<=(n-line+1); star++){
                Sop("*");
        }
        S.o.pln();
    }
}
```

→ Output :- Enter the value : 4

```
* * * *
* * *
* *
*
```

□ Print HALF - Pyramid Pattern

```
1
1 2
1 2 3
1 2 3 4
```

① outer loop → 4 times
② inner
③ number (inner loop count)

& Code :-

```
import java.util.Scanner;

public class patterns{
```

```
public static void main (String [] args){
        int n;
        Scanner sc = new Scanner (System.in);
        S.o.p ("Enter the value :");
L=1  N = x 2
L=2  N = x x 3        n = sc.nextInt ();
L=3  N = x x x
L=4  N = x x x x      for (int line = 1; line <= n; line ++) {
L=5  N = x x x x x 6
L=6  N = x            for (int no = 1; no <= line ; no ++) {
                          Sop (no);
                      }
                  Sopln();
              }
          }
      }
```

→ Output :- Enter the Value : 5

```
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
```

Print character Pattern

A
B C
D E F
G H I J

① outer n=4
② inner loop
③ Print
char ch = 'A', 'B', 'C'

```
for(int line = 1 to n)
    for (int char = 1 to line)
        print (ch)
        ch++;
```

Code:

```
import java.util.*;
public class patterns{

  public static void main (String [] a){

      int n;

      Scanner vj = new Scanner (System.in);
      Sopln (" Enter the Value of n: ");

      n = vj.nextInt ();
      Char ch = 'A';
```

```
for(int line =1; line <=n; line ++){
    for(int char =1; char <= line ; char++){
        S.o.p(ch);
        ch++;
    }
    Sopln();
  }
}
}
```

Dry Run

L=1	j= ~~1~~ 2	ch='A' B
L=2	j=~~1~~ ~~2~~ 3	ch=B C
L=3	j=~~1~~ ~~2~~ ~~3~~ 4	ch=D E F
L=4	j=~~1~~ ~~2~~ ~~3~~ ~~4~~ 5	ch=G H I J
L=5	j=~~1~~ ~~2~~ ~~3~~ ~~4~~ ~~5~~ 6	ch=K L M N O
L=6	j=X	ch=X

→ Output:- Enter the value of n: 5

```
A
B C
D E F
G H I J
K L M N O
```